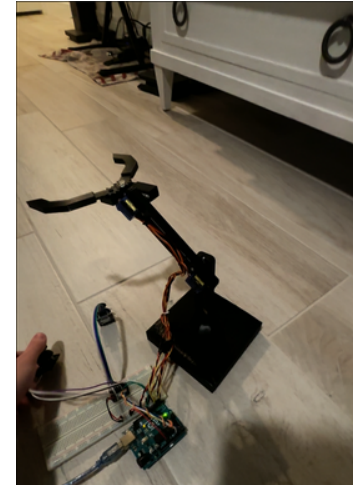
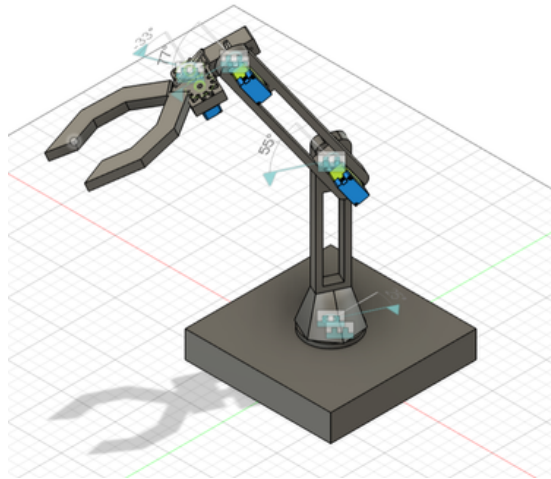
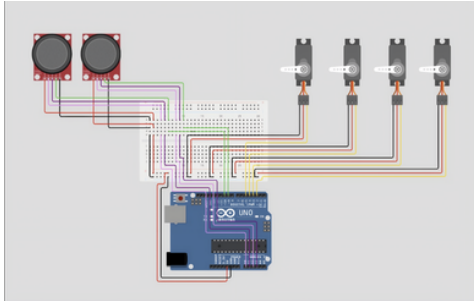




3 DOF ROBOTIC ARM - PERSONAL PROJECT



What?

- A **3D printed** robotic arm capable of precise object handling in any xyz coordinate
- Master **tolerances**, **servo integration**, and **DFA** principles through trial and error

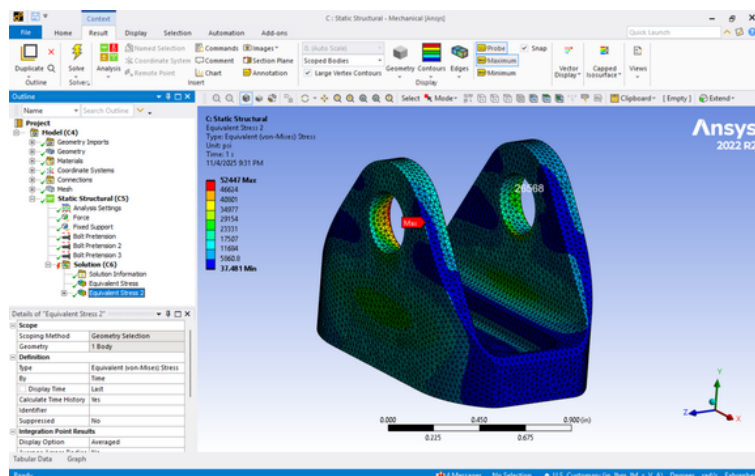
How?

- Applied H7/k6 tolerance to achieve **push-fits** when needed
- Implemented code and circuitry on **Arduino** after testing in TinkerCAD
- Holes designed to fit McMaster screws

Results

- Successfully **prototyped in PLA**, validated that mechanical tolerances and hardware-software interactions were appropriate

FRONT CLEVIS - GATOR MOTORSPORTS FSAE



What?

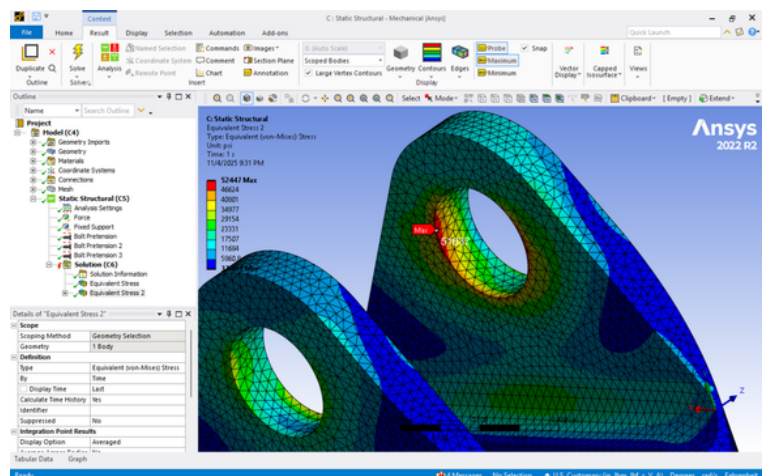
- Re-design front clevis for Gator Motorsports FSAE 2026 car
- Perform **static structural analysis** to optimize stress and strain on aluminum 6061-T6

How?

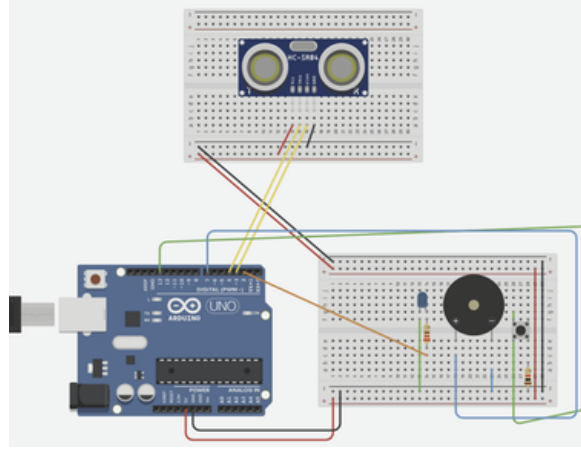
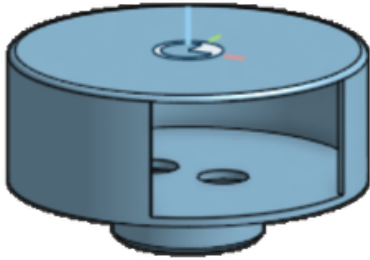
- Designed on **SolidWorks**
- Utilized **ANSYS mechanical** for **FEA** to analyze stress and strain concentrations
- Redesigned geometry at areas of high and low stress to improve part lifespan and minimize weight

Results

- Reduced weight on new clevis by approximately 8%
- Implemented **DFM** principles to reduce overall part production cost



HYDROALERT - SCHOOL PROJECT



What?

- Create a sensor-integrated water bottle lid to promote hydration and independent living for older adults with mild cognitive decline
- Integrate an ultrasonic sensor to monitor water-level, a buzzer & LED as the alert framework, and a button to reset alarm

How?

- Produced **3D CAD** model and detailed **ANSI-standard engineering drawings**
- Prototyped hardware and software for **Arduino** using TinkerCAD
- Fabricated using **rapid prototyping** methods (3D Printing)

Results

- Built successful prototype that triggers alert system when the bottle must be refilled or when the user has not drank water
- Reinforced **design principles** and **prototyping skills** after many design iterations